

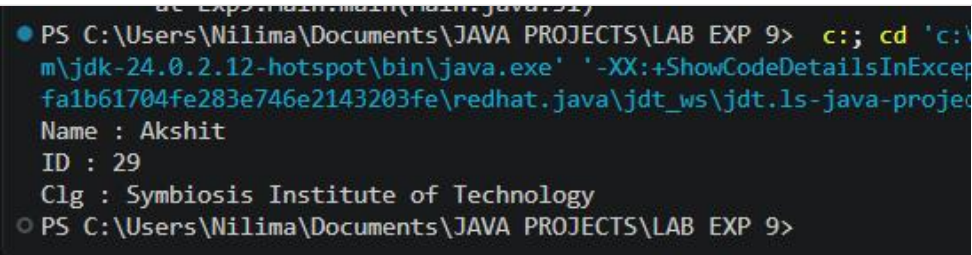


Symbiosis Institute of Technology, Pune

CSE- Academic Year 2026-27

Java Programming – Lab Batch 2025-29

Lab Assignment No.: 9	
Name of Student	Akshit Gawade
PRN No.	25070122029
Batch	2025-2029
Class	A1
Academic Year & Semester	Second Year Semester III
Date of Submission	11/7/2026
Title of Assignment:	Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.
Problem Statement	Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.
Learning Objectives	To implement the final keyword with variables, methods, and classes.
Theory	The final keyword restricts modification in Java programs. A final variable cannot be reassigned after initialization. A final method cannot be overridden and a final class cannot be inherited.
Program Code	Main:

	<pre> 1 final class College { 2 void collegeName() { 3 System.out.println(x: "Clg : Symbiosis Institute of Technology"); 4 } 5 } 6 7 class Person { 8 final int id = 29; 9 10 final void displayId() { 11 System.out.println("ID : " + id); 12 } 13 } 14 15 class Student extends Person { 16 String name; 17 18 Student(String name) { 19 this.name = name; 20 } 21 22 void displayStudent() { 23 System.out.println("Name : " + name); 24 } 25 } 26 27 public class Main { 28 Run Debug 29 public static void main(String[] args) { 30 Student s = new Student(name: "Akshit"); 31 32 s.displayStudent(); 33 s.displayId(); 34 35 College c = new College(); 36 c.collegeName(); 37 } 38 } </pre>
Output Screenshots	 <pre> PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 9> c:; cd 'c:\m\jdk-24.0.2.12-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionStack' Name : Akshit ID : 29 Clg : Symbiosis Institute of Technology PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 9> </pre>
Exercise Qns And Program Code	Q1. Create a Bank Account program where account number is final and cannot be changed.

```

1
2 class BankAccount {
3     final int accountNumber = 29;
4     String accountHolder;
5     double balance;
6
7     BankAccount(String accountHolder, double balance) {
8         this.accountHolder = accountHolder;
9         this.balance = balance;
10    }
11
12    void display() {
13        System.out.println("Account ID : " + accountNumber);
14        System.out.println("Account Holder Name: " + accountHolder);
15        System.out.println("Balance: " + balance);
16    }
17 }
18
19 public class Ex1 {
20     public static void main(String[] args) {
21         BankAccount b = new BankAccount(accountHolder: "Akshit", balance: 10500);
22         b.display();
23     }
24 }

```

Q2. Create a Library Book Management program where the book ISBN is declared as final and cannot be changed once assigned. Display the book's ISBN, title, author, and price.

```

1 class Book {
2     final String isbn;
3     String title;
4     String author;
5     double price;
6
7     Book(String isbn, String title, String author, double price) {
8         this.isbn = isbn;
9         this.title = title;
10        this.author = author;
11        this.price = price;
12    }
13
14    void display() {
15        System.out.println("ISBN : " + isbn);
16        System.out.println("Title : " + title);
17        System.out.println("Author : " + author);
18        System.out.println("Price : " + price);
19    }
20 }
21
22 public class Ex2 {
23     public static void main(String[] args) {
24         Book b = new Book(isbn: "978-93-12345-67-8", title: "Java Programming for Beginners", author: "R. Srinivasan", price: 649);
25         b.display();
26     }
27 }

```

Output Screenshots

```

● PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 9> c::; cd 'c:
Eclipse Adoptium\jdk-24.0.2.12-hotspot\bin\java.exe' '-XX:+ShowCo
ode\User\workspaceStorage\5e1afa1b61704fe283e746e2143203fe\redhat
Account ID : 29
Account Holder Name: Akshit
Balance: 10500.0
○ PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 9>

```

	<pre> ● PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 9> c:; cd 'c: Eclipse Adoptium\jdk-24.0.2.12-hotspot\bin\java.exe' '-XX:+ShowCo ode\User\workspaceStorage\5e1afa1b61704fe283e746e2143203fe\redhat ISBN : 978-93-12345-67-8 Title : Java Programming for Beginners Author : R. Srinivasan Price : 649.0 ○ PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 9> </pre>
Conclusion	This assignment helped in understanding the concepts of inheritance and interfaces in Java through real-world examples. It demonstrated how inheritance enables code reusability by allowing child classes to inherit the properties and methods of a parent class, while interfaces provide abstraction by defining common behaviors that can be implemented by different classes. The implementation improved my understanding of object-oriented programming principles such as reusability, abstraction, and polymorphism. These concepts are widely used in developing desktop applications, web applications, Android applications, enterprise software, e-commerce systems, banking systems, and other real-world Java applications that require modular, flexible, and maintainable code.